WEEK 8 · LECTURE L3 · UNIT III

KPI Frameworks and Building a Data Culture in HR

How measures cascade from strategy, why HR so often fails to use the analytics it has, and how to grow a data culture.

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DSEH001 · Digital Transformation in Human Resource Systems · M.Com. PGCF 2025 · Exam preparation edition · Updated 25 October 2026

AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

1. Apply the balanced scorecard, strategy maps and the HR scorecard, and cascade KPIs from strategy to actions.
2. Explain why HR struggles with analytics (Angrave et al.), and the failure modes of analytics programs.
3. Build a data culture with Schein, Ajzen, Argyris and Schön, stewardship and the change models.
4. Design an analytics function: structures, roles, governance, and a real human review.

ONE SOURCE

The same theories as the in class lecture and the course hub, without the simulation data used in class. Course outcomes: CLO 4 and CLO 3.

COMMIT BEFORE THEORY

Commit to an answer first

An HR team reports 40 measures every month, mostly cost and activity. The CEO says HR “measures everything and explains nothing”. What is the deepest fix?

A

Add more measures to cover strategy

B

Derive a few measures from the firm’s strategy, so each shows how people drive results

C

Report quarterly instead of monthly

D

Buy a better dashboard tool

EXAM LENS

The answer is B.

KAPLAN AND NORTON (1992; 2004)

The balanced scorecard and strategy maps

Financial

How do we look to shareholders?

Customer

How do customers see us?

Internal business process

What must we excel at?

Innovation and learning

Can we continue to improve and create value? People measures sit here and drive the other three.

EXAM LENS

Financial measures alone report the past; a balanced set adds the drivers of future performance. A strategy map (2004) draws cause and effect from learning and growth to processes, customers and finance: better managers, lower attrition, better service, sales. The arrows are hypotheses organizations rarely test; test each with data.

BECKER, HUSELID AND ULRICH (2001)

The HR scorecard

Doables

Activities HR performs: courses run, posts filled.

Deliverables

Outcomes that drive strategy: the right people, capable and committed, in key roles.

Alignment

HR practices that fit each other and the strategy.

Efficiency

Cost control, kept in balance with value created.

EXAM LENS

Measure the deliverables that drive strategy, not the doables HR finds easy to count.

CASCADING

From strategy to a measure someone can act on

Step	Example
Strategy	Grow service revenue in small towns
Business driver	Customers served by skilled, stable staff
HR deliverable	A stable, capable frontline in small town branches
KPI	First year retention of frontline staff, by town tier, with owner and target

Level	Measure	Action it supports
CHRO	Organization wide attrition (lagging)	Set priorities and budget
Regional HR partner	Share of staff under weakly rated managers (leading)	Choose which managers to coach
Line manager	My team’s manager effectiveness score	Change how I run my team

EXAM LENS

Each KPI should trace back to strategy in two or three steps; pair each lagging outcome with a leading driver someone can act on.

ANGRAVE, CHARLWOOD, KIRKPATRICK, LAWRENCE AND STUART (2016)

Why HR struggles to use analytics

Weak analytical capability

HR teams often lack the skills to frame and test questions.

Vendor led metrics

Tools report standard HR metrics, not the firm’s strategic questions.

Measures without strategy

Data collected because it is available, not because it matters.

Others may take over

If HR stays out, finance or IT will model people only as a cost.

EXAM LENS

Their remedies: strategic understanding of how people create value in this business; theory driven analytics grounded in how data were generated; focus on pivotal segments; experiments and quasi experiments to test cause.

WHY ANALYTICS PROGRAMS FAIL

Seven failure modes

1 to 3

HR skills gaps; a culture that privileges intuition; poor data quality and integration (headcount differs by system).

4 to 6

Unclear ownership and governance; short term performance pressure; weak executive sponsorship.

7 · The last mile

Insights exist but never enter daily decisions.

RECENT EVIDENCE · FERNANDEZ AND GALLARDO-GALLARDO (2021)

A review of 64 studies from 2010 to 2019 identifies 14 barriers to HR analytics adoption in four groups: data and models, software and technology, people, and management.

Course’s Data Storytelling and Change Enablement teaching note.

THEORY IN DEPTH · SCHEIN (1985)

Culture has three levels

The core idea

Artifacts (what you see), espoused values (what people say), basic underlying assumptions (what is taken for granted).

How it works

Change at the artifact level, such as a new dashboard, fails if assumptions stay the same.

Limits and critique

Assumptions are hard to observe, so diagnosis relies on interpretation.

IN PRACTICE

If line managers assume “HR data is for HR”, no dashboard will change their decisions.

EXAM LENS

Use Schein to explain why a data culture cannot be installed with tools.

THEORY IN DEPTH · AJZEN (1991)

Intention to use analytics has three roots

The core idea

The theory of planned behavior: attitude, subjective norms and perceived behavioral control predict intention.

How it works

People use analytics when they value it, when leaders expect it, and when they feel able to.

Limits and critique

Intentions do not always become behavior.

IN PRACTICE

Leaders asking for data at every review builds the norm; training builds perceived control. Training alone fails if attitude and norm are unchanged.

EXAM LENS

Map each data culture intervention to one of Ajzen’s three roots.

ARGYRIS AND SCHÖN (1978)

Single and double loop learning

Single loop

Are we hitting the target? Adjust actions within existing goals: turnover rises, so raise retention bonuses. Analytics as a thermostat.

Double loop

Is this the right target? Question goals, norms and assumptions: is money really the driver? Analytics as a mirror.

EXAM LENS

A data culture needs both. Analytics enables double loop learning only with psychological safety, transparent data and a habit of experiment.

OWNERSHIP AND SPREAD

Stewardship, cadence and adopters

Stewardship (Davis, Schoorman and Donaldson, 1997)

People act as stewards of organizational interests when they have ownership, authority and regular feedback. A metric without an owner is no one’s responsibility.

Cadence

Weekly for operational metrics, monthly for tactical, quarterly for strategic.

Spread (Rogers)

Start with innovators and early adopters to build proof; do not spend early effort on laggards.

CHANGE MODELS FOR ANALYTICS

Kotter, ADKAR and Lewin applied

Kotter

Urgency from the real cost of poor decisions; a coalition of HR, HRIS, business users and skeptics; champions in every unit; visible wins; analytics built into planning and talent reviews; roles and governance that outlast champions.

ADKAR

Gaps usually sit at Desire (resistance to data led decisions) and Ability (training without practice, tools or coaching).

Lewin

Unfreeze by showing the limits of intuition; change with intensive support; refreeze through competencies in appraisals, governance policies and norms that reward rigor.

DAVENPORT AND HARRIS (2007; 2017)

Building the analytics function

Analytical competitors

Senior sponsorship, enterprise wide data and analysis embedded in core decisions, reached in stages from localized to enterprise analytics. For most HR functions, step one is a named owner with the CHRO’s sponsorship.

Structures

Centralized (consistent, but a bottleneck); decentralized (close, but inconsistent); hub and spoke (center of excellence plus embedded partners); federated (councils and communities).

Roles and governance

Analytics leader, analysts, translators who turn findings into decisions (often the missing role), engineers, visualization specialists, change managers. Five governance domains: data, prioritization, ethics and risk, standards, communication.

A REAL HUMAN REVIEW

Who, how often, by what criteria, and escalation

Who reviews

A named role with authority to overturn the algorithm.

How often

A set cadence, not only when someone complains.

By what criteria

Written standards for when to override.

Escalation

Where a disputed case goes next, and how fast.

EXAM LENS

A review without all four is a rubber stamp.

This four part specification is the course’s own framing.

BUILDING DATA LITERACY IN HR

Five capabilities

Ask

Turn a business problem into an answerable question.

Read

Interpret rates, comparisons and uncertainty correctly.

Challenge

Spot vanity numbers, bad frames and weak causal claims.

Decide

Use evidence alongside judgment and stakeholder values.

Own

Name an owner and a review rhythm for each KPI.

RECENT EVIDENCE · MCCARTNEY, MURPHY AND MCCARTHY (2021)

A competency model for HR analysts, built from 110 job advertisements in five countries and 12 interviews, shows what employers actually ask for.

Course framing: literacy is the capability Angrave et al. found missing; tools cannot supply it.

APPLY

Forty measures and no story

Teaching case (Course framing)

A Jaipur based microfinance company plans to double its rural branches in three years. Its HR report lists 40 monthly measures: training hours, posts filled, cost per hire, leave taken. Branch staff turnover is 35%. The CEO asks HR which three numbers tell him whether people will make the expansion work.

Work it through

1. Sort typical measures into doables and deliverables.
2. Cascade from the strategy to three KPIs.
3. Which leading indicator would warn early?
4. What single loop and double loop questions should HR ask?

COMMON CONFUSIONS

Tempting readings, and better ones

A tempting reading	A better reading
A good HR scorecard measures everything HR does	It measures the deliverables that drive strategy
The balanced scorecard is a finance tool	It balances results with their drivers, including people
Better tools create a data culture	Capability, assumptions and questioning habits create it
A human in the loop makes an algorithm safe	Only if who, how often, criteria and escalation are specified

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse (1 of 2)

Using the balanced scorecard, the HR scorecard and single and double loop learning, design a KPI framework for an Indian bank expanding into small towns, and explain how you would build the data culture to use it. **20 marks**

A strong answer: places people measures in the scorecard and shows the causal chain; distinguishes doables from deliverables and cascades three KPIs; includes leading indicators, owners and targets; applies single and double loop learning and Angrave et al. to the culture plan.

“You cannot install a data culture; you have to grow one.” Discuss using Schein, Ajzen and Davenport and Harris. **20 marks**

A strong answer: distinguishes artifacts from assumptions; maps interventions to the three roots; sequences realistically.

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse (2 of 2)

Design a cascading KPI framework for reducing frontline attrition, and evaluate the causal assumptions it rests on. **20 marks**

A strong answer: cascades across levels with leading and lagging measures and owners; critiques strategy map arrows; proposes tests.

Why, according to Angrave et al. (2016), is HR set to fail the big data challenge? **10 marks**

A strong answer: states their argument and proposed remedies.

CARRY THESE INTO THE EXAMINATION

Four things to remember

1. **Derive a few KPIs from strategy.**
   Deliverables, not doables.

2. **Cascade every KPI to a measure someone can act on.**
   With an owner and a cadence.

3. **Culture changes at the level of assumptions.**
   Not dashboards.

4. **A real human review names who, how often, criteria and escalation.**
   Otherwise it is a rubber stamp.

SET READINGS · FROM THE LESSON PLAN

Read for this lecture

- Johnson, Carlson and Kavanagh (2025), Chapter 11; Angrave et al. (2016).

NEXT WEEK

- Tambe, Cappelli and Yakubovich (2019); Kellogg, Valentine and Christin (2020); Leicht-Deobald et al. (2019).

> **NOTE**
> The other works named in this lecture are theory sources for reference, not set readings.

Sources for this lecture

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